

Swesan Pathmanathan

Canadian Citizen

+1-647-447-9580 | 201360on50@gmail.com | linkedin.com/in/swesan | github.com/swesan123

EDUCATION

McMaster University

Bachelor of Engineering, Software Engineering (Co-op)

Hamilton, ON

Sep 2021 – Apr 2026

TECHNICAL SKILLS

Languages: C, Python

Systems: Linux, Windows, Bash

Data: NumPy, scikit-learn

WORK EXPERIENCE

Advanced Micro Devices (AMD) | Python, Linux, Windows

Datacenter GPU Infrastructure & Validation Engineer (Co-op)

Markham, ON

May 2024 – Aug 2025

- Executed post-silicon functional validation and integration test passes for MI300X, MI325X, and MI350X, covering performance, IO interfaces, and platform quality across 100+ runs per week.
- Built **Linux** and **Windows** GPU validation configurations with varied BMC, BIOS, and OS images to stress RAS, resets, IO, performance, and system management, reducing bring-up issues by about 20%.
- Automated regression execution with **Python** scripts and an internal workload launcher so validation suites ran faster, cutting manual setup from about 20 minutes to under 5 minutes.
- Triaged post-silicon failures across ROCm, power-management firmware, drivers, and hardware interfaces using logs and reproducible steps, shortening time-to-root-cause on assigned issues by roughly 30%.
- Supported firmware and validation teammates via Jira and updated written checklists and notes so less experienced engineers could repeat complex platform validation setups with fewer misconfigurations.

PROJECTS

ALU Design with Sequential Circuits | Verilog, Quartus

September 2025

- Built an **Arithmetic Logic Unit (ALU)** in **Verilog** using simple, synthesizable modules, passing all planned functional tests.
- Used **Quartus** waveform views and testbenches to check that arithmetic and logic operations behaved as expected across dozens of input cases.

SufSort Sorting System | C, Bash, Linux

November 2025

- Wrote two sorting programs in **C** to compare how different algorithms behave on large data files of up to several hundred megabytes.
- Used **Bash** scripts on **Linux** to generate input data, run batch tests, and check sorted output, catching logic errors early in development.

Linux Log Corpus Analysis | Python, scikit-learn, NumPy

February 2026

- Built a simple analysis pipeline in **Python** to read and process tens of thousands of labeled Linux system logs.
- Used **NumPy** and **scikit-learn** to find which log terms best separate different log categories, improving signal-to-noise for downstream analysis.

RELEVANT COURSES

Real-Time Systems, Digital Systems Interfacing, Computer Architecture